Tamanna Bhrigunath

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SUMMARY

B.Tech graduate in Artificial Intelligence & Data Science (2026) with hands-on experience building and deploying production ERP modules and full-stack web applications. Skilled in Python, Flask, REST APIs, SQL, PostgreSQL, Next.js, and database design. Previously developed attendance, biometric verification, and payroll features during a software engineering internship, alongside end-to-end machine learning projects. Interested in building reliable backend systems and scalable software products.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, TypeScript
- **Backend & Web:** Flask, REST APIs, SQLAlchemy, Flask-Login, Next.js, React, Prisma, HTML, Tailwind CSS
- **Databases:** PostgreSQL, MySQL
- **APIs & Auth:** Google Calendar API, OAuth 2.0, TMDB API
- **Deployment & Tools:** Render, Vercel, Gunicorn, Git, VS Code, Jupyter
- **Machine Learning:** scikit-learn, pandas, NumPy, Regression, Random Forest, SVM, Feature Engineering, Model Evaluation

EXPERIENCE

Software Engineering Intern — Nessco Paper Cup Machine Pvt. Ltd. *Jan 2026 – Aug 2026*
Jaipur, India

- Built and shipped 3 production ERP modules — attendance tracking, biometric face check-in, and payroll — used daily by company staff.
- Designed a race-safe attendance punch system using idempotency keys and a [userId, date] unique constraint, eliminating duplicate records under simultaneous retries.
- Implemented browser-side face verification (face-api.js) with 3-frame passive liveness detection and server-side Euclidean-distance matching, keeping raw biometric data off the server.
- Built a real-time payroll engine (Next.js, Prisma, PostgreSQL) recomputing payslips on every read — covering PF, ESI, PT, TDS, and overtime — removing the need for nightly batch jobs.

Data Science Intern — Celebal Technologies *Jun 2025 – Aug 2025*
Remote

- Shipped 2 end-to-end ML pipelines from raw data to deployed Streamlit apps: a house price predictor and a credit risk classifier.
- Built a credit risk classifier (Random Forest, German Credit dataset); resolved class imbalance via resampling, reaching ~80% accuracy evaluated on precision, recall, and F1.
- Benchmarked Linear Regression, Random Forest, and SVM via cross-validation for house price prediction, selecting the best-generalizing model.

PROJECTS

HiveFlow — Collaborative Task Manager *GitHub · Live*
Flask · SQLAlchemy · OAuth 2.0 · PostgreSQL · Render

- Designed and deployed a multi-tenant task management platform with organizations, Kanban boards, role-based access, CSV export, and email OTP password recovery.
- Integrated Google Calendar API (OAuth 2.0) to auto-sync task creation, updates, and deletions to users' calendars, handling token refresh and API failures gracefully.
- Modeled relational data in SQLAlchemy with strict user-level task isolation; deployed on Render with Gunicorn and PostgreSQL.

Movie Recommendation Engine *GitHub · Live*
scikit-learn · CountVectorizer · TMDB API · Streamlit

- Built a content-based recommender combining genre, cast, director, and plot into a single feature vector, ranking top-5 similar titles by cosine similarity, with live posters via the TMDB API.

EDUCATION

B.Tech, Artificial Intelligence & Data Science — Poornima College of Engineering *2022 – 2026*
Jaipur, India

- Relevant Coursework: Machine Learning, Data Mining, DSA, Computer Networks, DBMS, Operating Systems, OOP
- Runner-up, AADHAR-XI Technical Project Competition (2023) — Zircon Club, Poornima College of Engineering